

From One Chart to a Full Financial Model

How the Rent vs. Buy Calculator Evolved

Every capability on the following slides started as a question — not a spec.

Where We Started

Most rent-vs-buy calculators only compare monthly payments. Build one that also asks what the down payment could have earned elsewhere.

- A 30-year net worth comparison, renter vs. buyer
- One clear answer: the **break-even year**

Hyper-Local Rent vs. Buy Opportunity Cost Calculator

Most calculators ignore the opportunity cost of your down payment. This one invests it in the stock market and tracks a true 30-year net worth comparison.

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THE PROPERTY

Zip Code (California)
e.g. 94110

Import from Zillow Screenshot

Home Price
\$650K

Monthly Rent (comparable)
\$2.8K

HOMEOWNERSHIP COSTS

Down Payment
20% (\$130K)

Mortgage Rate
6.5%

Property Tax Rate
1.10%

Monthly HOA
\$250

Annual Home Insurance
\$1.2K

BREAK-EVEN POINT
Year 15

Buying overtakes renting's investment returns in year 15.

Mortgage (P&I) / mo
\$3,287

Buyer Net Worth (Yr 30)
\$1,520,423

Renter Net Worth (Yr 30)
\$1,550,453

30-Year Net Worth Projection

Deterministic Monte Carlo

→ Buying (Home Equity) → Renting (Stock Portfolio)

THE QUESTION

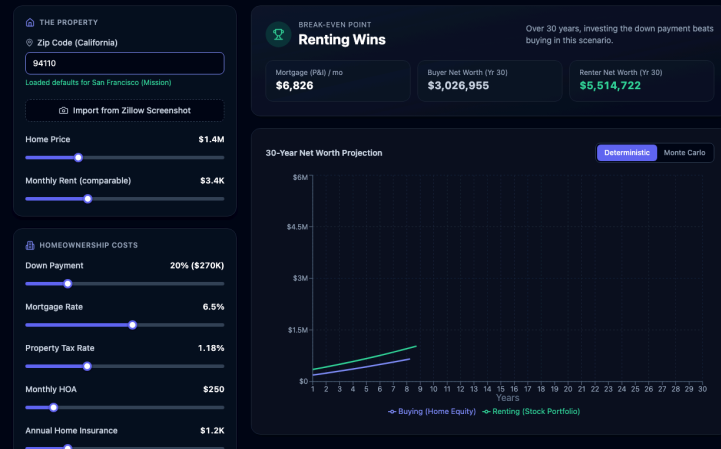
"How can we make this feel local, not generic?"

Zip-code-aware defaults

Type a zip code, get real market numbers instantly — home price, rent, and property tax rate — instead of guessing at three unfamiliar sliders.

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THE QUESTION

"What costs are we missing?"

Insurance and maintenance, modeled properly

Home insurance and maintenance don't inflate at the same pace as rent, so they got their own sliders and their own inflation rates — not lumped together as one vague "cost of owning" number.

The section was renamed from "Mortgage & Costs" to "**Homeownership Costs**" to match what it actually covered by this point.

THE QUESTION

"A single guess isn't good enough — what about risk?"

A second view: Monte Carlo simulation

Markets don't move in a straight line. A toggleable second view runs 500 simulated 30-year outcomes and shows a range, not false precision.

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BREAK-EVEN POINT (MEDIAN)

Year 11

In the median simulated scenario, buying overtakes renting in year 11.

Mortgage (P&I) / mo

\$3,287

Buyer Net Worth (Yr 30), median

\$1,484,338

Renter Net Worth (Yr 30), median

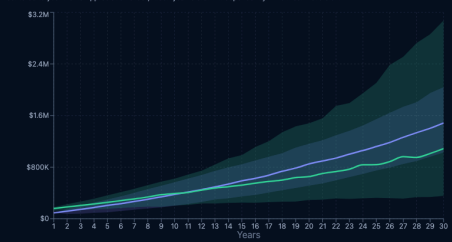
\$1,086,328

30-Year Net Worth Projection

Deterministic

Monte Carlo

Buying wins in **85%** of 500 simulated 30-year scenarios. Shaded bands show the 10th-90th percentile range. The renter's returns are bootstrapped from 1928-2025 S&P 500 annual total returns (NYU Stern) (re-centered to your Stock Market Return slider), home appreciation is bootstrapped from the FHFD Case-Shiller U.S. National Home Price Index, 1987-2025 (re-centered to your Home Appreciation slider). Each year is drawn independently — no mean-reversion.



THE QUESTION

"Can that simulation be grounded in real history, not a guess?"

Real market history, not invented statistics

The simulation originally sampled from a made-up statistical curve. It now resamples from what actually happened:

- Real S&P 500 annual returns, **1928–2025**
- The real Case-Shiller national home price index, **1987–2025**

Every simulated year is a year that genuinely occurred, just reshuffled.

THE QUESTION

"What about taxes — isn't that the biggest lever?"

Modeling what actually shows up on a tax return

- Mortgage interest and property tax deductions, weighed against the standard deduction
- Capital gains tax on the renter's investment gains
- The home-sale tax exclusion buyers get — \$250K or \$500K tax-free — which most calculators ignore entirely

THE QUESTION

"Taxes got complicated — can we make sense of it?"

Giving taxes their own home

Every tax-related input moved into one dedicated section. A state tax rate was added too, since California taxes investment gains as ordinary income — another detail most calculators miss.

THE QUESTION

"How do I share exactly what I'm looking at?"

One click, one link

Every input and every setting — encoded into a single URL. No accounts, no saved files, no re-explaining your numbers to someone else.

THE QUESTION

"Do I have to type in every number by hand?"

Upload a screenshot instead

Drop in a screenshot of a listing and the key numbers get pulled out automatically. Nothing is applied silently — every extracted value is shown for review and confirmation first.

THE QUESTION

"Can I take this analysis with me?"

A real report, one click away

Your exact scenario — every chart, every input, every assumption — turns into a clean, shareable document instead of living only on screen.

THE QUESTION

"What if I don't want to bet on the stock market?"

Choose your own investment

Stocks, Treasury bonds, or gold — each with its own real historical data behind it. Same rigor, whichever path you're actually weighing.

Every Feature Started With a Question

None of this was planned upfront. Each capability came from asking "can we also..." — and the model just kept getting more honest.

onevvish.github.io/rentVsbuy